

Extra fourth-order modes: compact currents and black-hole causal tests

Bridge note for conformal-gravity, higher-derivative quantization, and Einstein-from-conformal researchers. Status: compact classical current theorem and scoped Schwarzschild axial causal-disposition theorem, 19 July 2026.

1. Their object and the unresolved question

Three adjacent programmes ask different questions. Maldacena uses a Neumann boundary condition in asymptotically de Sitter or Euclidean anti-de Sitter settings to select the Einstein solution in a tree-level computation (arXiv:1105.5632). Mannheim argues for a positive PT -based inner product in higher-derivative gravity (arXiv:2109.12743). Kubo and Kuntz use a BRST/Fock construction and find an indefinite transverse spin-two sector and scattering-unitarity failure (arXiv:2202.08298). These results concern different boundaries, state spaces, and inner products.

Our narrower classical question is prior to all three quantum conclusions: on one compact common Einstein–Maxwell/Weyl–Maxwell background, are the extra fourth-order roots actual gauge-quotient solution directions, are they radical under the action-derived current, and where does the negative classical current direction lie? The axial block now answers the full current question; the polar block supplies an independent off-shell equation-module comparison whose direct current remains the next gate. A second, independent Schwarzschild laboratory now asks whether horizon regularity and causal decay can select the Einstein branch. In its first axial fixture, they cannot.

```
H^0(Einstein--Maxwell)  --injects-->  H^0(Weyl--Maxwell)
      |                               |
      | standard image                +--> Q_extra = two axial directions
      |                               +--> two polar equation directions
      |                               |
      +----- direct Lee--Wald matrix -----+
                    image signs (1,1), extra signs (2,0)
                    full axial signature (3,1)
```

2. Exact dictionary

Item	Adjacent usage	This project	Conversion or caveat
Theory	pure conformal gravity or Einstein branch	tuned Weyl–Maxwell versus Einstein–Maxwell	Matter and compact flux are part of the fixture.

Item	Adjacent usage	This project	Conversion or caveat
Background	dS/EAdS, flat Fock vacuum, or higher-derivative vacuum	compact $\mathbb{R} \times S^1 \times S^2$ product with fixed $U(1)$ bundle; separately, Schwarzschild exterior	Results are compared, not identified, across these two laboratories.
Extra mode	fourth-order pole/Jordan/Fock excitation	quotient $Q_{\text{extra}} = H_{WM}^0/i_*H_{EM}^0$	Quotient solution directions are not yet particles.
Gauge	BRST physical-state quotient or boundary selection	local $\text{Diff} \times \text{Weyl} \times U(1)$, before final residual quotient	Boundary selection is not identified with gauge quotienting.
Form	Dirac, PT , or Fock inner product	classical covariant Lee–Wald current	A classical sign is not a quantum norm.
Selection	boundary condition or quartet mechanism	compact harmonic regularity and fixed bundle; local horizon regularity and asymptotic decay on Schwarzschild	The axial Schwarzschild local causal test is now negative; general scattering selection remains open.

3. Reproduced benchmark

The current engine first reproduces the independent Einstein–Maxwell Lee–Wald current, the repository’s Bach convention, a pure-Weyl gauge kernel, the flat transverse-traceless zero-restriction control, and exact current conservation. It then compares the reduced self-adjoint Hessian current with a literal variation of the four-dimensional Weyl–Maxwell curvature-momentum current. For arbitrary $\ell \geq 2$ and spherical multiplicity m , their integrated currents agree up to the positive harmonic normalization $N_{\ell m}$.

The replay commands are:

```
python3 -m bridge.einstein_sector.einstein_maxwell_weyl_axial_lee_wald_completion \
--verify bridge/certificates/einstein_maxwell_weyl_axial_lee_wald_completion.json
python3 -m bridge.einstein_sector.weyl_maxwell_axial_general_lee_wald_fixture \
```

```

--verify bridge/certificates/weyl_maxwell_axial_general_lee_wald_fixture.json
python3 -m bridge.einstein_sector.einstein_maxwell_weyl_polar_full_tensor \
--verify bridge/certificates/einstein_maxwell_weyl_polar_full_tensor.json
python3 bridge/einstein_sector/verify_einstein_maxwell_weyl_polar_full_tensor.py
python3 -m bridge.einstein_sector.einstein_maxwell_weyl_polar_physical_completion \
--verify bridge/certificates/einstein_maxwell_weyl_polar_physical_completion.json
python3 bridge/einstein_sector/verify_einstein_maxwell_weyl_polar_physical_completion.py
python3 black_hole_programme/verify_bh2a_cross_flux.py
python3 black_hole_programme/verify_bh2a_causal_disposition.py
python3 black_hole_programme/verify_bh2b_polar_split.py

```

4. Added result

Generic compact axial Lee–Wald theorem. For each physical compact axial harmonic with $\ell \geq 2$, the Weyl–Maxwell solution module strictly contains the Einstein–Maxwell image by two exact polarizations. The direct four-dimensional Lee–Wald form is nondegenerate on the extra quotient with signature $(2, 0)$. The complete generic axial target has signature $(3, 1)$, and its negative direction lies on one Einstein-image master branch rather than on either extra direction.

Algebraically, the invariant factors separate the standard Einstein master polynomial from a doubled extra primary factor. Direct mixed Lee–Wald blocks between the standard and extra primary shells vanish. The extra determinant is positive throughout the physical $\ell \geq 2$ domain. This rules out three simple explanations of the extra root on this fixture: determinant multiplicity only, local gauge, and a presymplectic radical.

The nonlinear compact test adds an important qualification. The real $\ell = 2, k = 0$ extra span has a definite Taub obstruction at fixed bundle charges, so the linearly measurable extra directions are linearization unstable in that declared sector. This is selection by nonlinear constraint, not disappearance from the linear current.

The independent polar calculation now establishes the corresponding equation-level decomposition. A formally self-adjoint four-by-four target Hessian obeys the exact polynomial Einstein square

$$H_P S_P = J_P E_P,$$

without dividing by momentum or either characteristic factor. On every declared physical $\ell \geq 2$ compact-momentum fibre, including $k = 0$, its invariant factors are $1, 1, p, pq$. The Einstein image is the complete q -primary summand, the canonical extra quotient is $(K[\omega]/(p))^2$, and the action row weights $(-1, 2, -1, 2\lambda)$ are derived from the four-dimensional variation and harmonic

norms. This is not yet a polar current theorem: the direct polar Lee–Wald form, ungauged BV/Noether lift, and residual descent remain open.

5. Independent Schwarzschild boundary test

The black-hole programme supplies a second laboratory in which boundary admissibility is a physical part of the question rather than a compact-mode regularity convention. For Schwarzschild mass $m = 1$, axial $\ell = 2$, and the declared real-frequency ingoing family, it certifies three successive facts:

1. the Einstein/Regge–Wheeler block is symplectically null in the pure-Weyl action current for conjugate wave pairs;
2. the Einstein–extra and extra–extra horizon-flux blocks are nonzero at the exact fixtures $\omega = 3/5, 2/7$, and the independent verifier fixture $1/2$;
3. the extra branch is horizon-regular, luminal and bounded at infinity with the same leading falloff as the Einstein branch.

Consequently, no ordinary local causal regularity or decay condition at either end removes the extra axial branch on this fixture. Selecting the Einstein branch alone would require an additional projection on scattering data that relates temporal ends, rather than a causal initial-boundary condition imposed locally. This is a scoped causal-disposition no-go. It does not establish instability, a quasinormal spectrum, a global scattering theory, or a quantum norm.

The companion polar audit has also begun. On an arbitrary Ricci-flat background the linear Bach equation splits exactly into the Einstein condition and a trace-coupled Lichnerowicz-type system for the extra Ricci curvature. Polar horizon series, action-derived flux, outer-boundary disposition, and stability remain open.

6. Consequence in their language

The result does not choose between PT , Fock–BRST, or boundary-selected quantization. It says that any comparison must account for an explicit classical carrier space in both parities. In the axial block, “extra branch” and “negative direction” are demonstrably not synonymous; the polar current will test whether that separation persists. A successful boundary or quantum prescription must explain which standard and extra compact directions survive its own quotient and why its inner product is the transported form appropriate to that state space.

For Einstein-from-conformal work, this is a Lorentzian/nonlinear complement: linear Einstein inclusion holds, but the identity inclusion is not symplectic and selected real tangents fail the second-order compact Taub test. That does not refute the EAdS/dS boundary theorem; it asks whether its selection is causally preserved and symplectically appropriate in real time.

The Schwarzschild result sharpens that question. A Maldacena-type selection

may still define a valid sector, but in this Lorentzian axial laboratory it cannot be implemented merely as ordinary horizon regularity plus causal decay at infinity. It must act as a genuine branch selection on scattering data. That distinction is potentially relevant whenever a Euclidean or two-boundary prescription is interpreted as a real-time causal truncation.

7. Scope boundary

The results are **LOCAL-ALGEBRAIC/REDUCED-MODE** on a compact product background. They are not a positive-frequency Hilbert norm, a BRST/Fock theorem, a *PT* metric comparison, a quantum ghost result, or an S-matrix statement. The polar extra current and ungauged lift, final residual quotient, literal second expansion of the four-dimensional action density, and causal/asymptotic boundary phase spaces remain open. The compact negative direction may be removed, retained, or reinterpreted by those later gates. The black-hole theorem is only axial $\ell = 2$, $m = 1$, real frequency and the declared horizon/asymptotic domains. It is not a result for polar modes, complex frequencies, arbitrary multipoles or masses, the full exterior initial-boundary problem, nonlinear stability, ringdown, Hawking states, or asymptotic particles.

8. One useful question for adjacent experts

Is there an admissible Lorentzian boundary or BRST selection that removes the negative Einstein-image master direction while treating the two positive extra directions consistently, and can that selection be expressed as a causal symplectic subquotient rather than a condition imposed at both temporal ends?

Reproducibility receipt. Sources: arXiv:1105.5632v2, arXiv:2109.12743v1, arXiv:2202.08298v2. Certificates: **AXIAL_LEE_WALD_COMPLETION**, **POLAR_PHYSICAL_COMPLETION**, **BH2A_CROSS_BLOCK_NONZERO_HORIZON_FLUX_FIXTURES**, and **BH2A_AXIAL_CAUSAL_DISPOSITION_EXTRA_BRANCHES**. Verification commands are in section 3. Tags: **LOCAL-ALGEBRAIC**, **REDUCED-MODE**, and the scoped black-hole result **LORENTZIAN-CAUSAL**. Open: polar current/lift, residual descent, full causal boundary phase spaces, stability, and quantum state.